

Section 1 Data Availability

The dataset used in this study is publicly available at https://github.com/jackie-illiilli/ML_Bioorthogonal_DI. The repository includes the following directories: *Data*, *Code*, and *Figure*. The *Code* directory contains scripts related to the construction of initial transition state guesses, data processing, model training, and the output results of AttentiveFP and QM-GNN. The *Data* directory stores critical information from the entire screening process, while the *Figure* directory contains the figures used in the article.

Details of the subdirectories within the Data directory are as follows: *ZINC_Data* includes the step-by-step results of screening from the ZINC dataset, with the final selected 4π and 2π candidates stored in the files *target_diene_passed.csv* and *target_ene_passed.csv*, respectively. *DFT_Result* contains the results of DFT calculations, where the reaction Gibbs free energy (ΔG_{rxn}) is stored in *Rxn.csv*, and the activation Gibbs free energy ($\Delta G^\ddagger$) along with Distortion/Interaction Energy results are stored in *Active_Energy.csv*. *Reported_BO* includes bioorthogonal reactions previously reported in the literature that we identified. *Predict_DI* stores the results of the screening process for potential bioorthogonal reactions, with *BO_LowThan_24_potential.csv* containing the identified potential bioorthogonal reactions.

Section 2 Details of Screening Representative DA Reactants and Dataset

Section 2.1 Screening DA Reaction Scaffolds from the ZINC Dataset

Reactants were systematically retrieved from the ZINC20 in-stock database (<https://zinc20.docking.org/>), a comprehensive resource of commercially available compounds provided by the Irwin and Shoichet Laboratories at the University of California, San Francisco (Irwin et al., J. Chem. Inf. Model., 2020, DOI: 10.1021/acs.jcim.0c00675). Molecules with a molecular weight of ≤ 300 Daltons were downloaded for subsequent evaluation. In the initial screening phase, only reactants with ≤ 10 heavy atoms and composed exclusively of H, C, N, O, and F elements were retained. This constraint was applied to minimize conformational diversity in the reactant pool while ensuring coverage of all known bioorthogonal reaction scaffolds identified to date. By limiting the number of heavy atoms to 10, the focus was directed toward the core skeletal structures of these molecules, facilitating a streamlined analysis of their reactivity patterns. Additionally, SMILES strings containing formal charges were excluded to maintain chemical neutrality and simplify downstream processing.

The identification of 4π and 2π systems, critical for evaluating Diels-Alder (DA) reactivity, was performed using the SMARTS expressions '*==**=*' (for 4π systems) and '*=*' (for 2π systems). To refine the dataset further, a specific subset of 4π systems known to be unreactive in DA reactions was excluded. These systems feature a conjugated arrangement with one double bond and an intervening single bond within a ring structure, paired with a second double bond positioned outside the same ring, rendering them incompatible with the stereoelectronic requirements of the DA reaction. Following these filtering steps, a total of 13,028 qualifying 4π reactants and 31,115 qualifying 2π reactants were retained for subsequent analysis.

Section 2.2 Screening Reactants with Structural Diversity

In the second phase of the workflow, molecular diversity screening was conducted to ensure a representative and structurally varied subset of reactants. This was achieved using the *MaxMinPicker* algorithm implemented in the *RDKit* module. Morgan molecular fingerprints were calculated with a radius of 4 and a bit vector length of 4096 to capture detailed structural features of the molecules. The similarity assessment relied on the default method of *MaxMinPicker.LazyBitVectorPick*, which iteratively selects compounds to maximize the minimum Tanimoto distance to previously picked molecules. To ensure reproducibility, a fixed random seed of 1 was applied consistently throughout the process. This diversity-driven approach was critical to avoid redundancy in the dataset, enabling the exploration of a broad chemical space while preserving the potential for novel reactivity patterns in downstream analyses. Following this step, a refined subset of 3,000 qualifying 4π reactants and 3,000 qualifying 2π reactants was retained for further investigation.

Section 2.3 Screening DA Reactants with Thermodynamic Feasibility

In the final step, Gibbs free energy barriers ($\Delta G^\ddagger$) were calculated for reactions involving all 4π systems with ethylene and all 2π systems with butadiene, providing a thermodynamic basis for assessing their suitability for Diels-Alder (DA) reactivity. This step was essential because a large proportion of reactants with 4π or 2π structures—particularly those exhibiting aromaticity—typically exhibit limited reactivity in bioorthogonal DA reactions due to their stabilized electronic configurations. However, notable exceptions exist, such as tetrazines, which possess aromatic character yet serve as highly effective 4π reagents in bioorthogonal applications. This duality necessitated a rigorous evaluation of each reactant's

thermodynamic fitness as a 4π or 2π component, rather than relying solely on simplistic functional group or aromaticity-based filtering. Consequently, only reactants with reaction free energy barriers of ≤ 10 kcal/mol were retained, yielding 718 qualifying 4π systems and 1,789 qualifying 2π systems. These compounds collectively defined a reaction space encompassing 1,284,502 unique reactant combinations for subsequent analysis.

Among these, the 4π systems encompass a diverse range of structures, including cyclic dienes ranging from four- to nine-membered rings, aromatic heterocyclic dienes bearing substituents with varied electronic effects, and conventional aliphatic dienes. The 2π systems include a broad selection of motifs commonly employed in bioorthogonal chemistry, such as strained cycloalkenes, olefins featuring carbon-heteroatom double bonds, alkynes, and aromatic hydrocarbons. A comprehensive list of all screened 4π and 2π reactants is provided in the **IndexN**.

Section 2.5 Screening of a DFT Accessible Dataset

To ensure robust model training, the DFT dataset must balance representativeness and diversity, thereby maximizing the predictive power of the model across the full reaction space while minimizing the required data volume. To this end, we utilized the MaxMinPicker tool from RDKit to select a subset of reactions for DFT calculations. Specifically, Morgan molecular fingerprints were generated for the 4π and 2π reactants using a radius of 2 and a bit vector length of 2048 (nBits=2048). These individual fingerprints were concatenated to form a 4096-dimensional fingerprint representing each reactant pair. This combined fingerprint was then processed using RDKit's LazyBitVectorPick method, which iteratively selects entries to maximize structural diversity based on Tanimoto distance. This screening yielded 5,000 thermodynamically accessible reactant combinations, forming the core DFT dataset. When accounting for site selectivity and stereoselectivity in the Diels-Alder (DA) reactions, these combinations correspond to over 50,000 distinct reaction profiles, providing a comprehensive foundation for subsequent modeling.

This set of 5,000 reactant combinations encompasses the complete pool of 718 qualifying 4π systems and 1,789 qualifying 2π systems identified earlier. To highlight the structural diversity within this thermodynamically accessible DFT dataset, we tabulated the most frequently occurring 4π and 2π structures. These representative examples underscore the broad chemical scope captured by the dataset, ensuring its utility for comprehensive reactivity modeling.

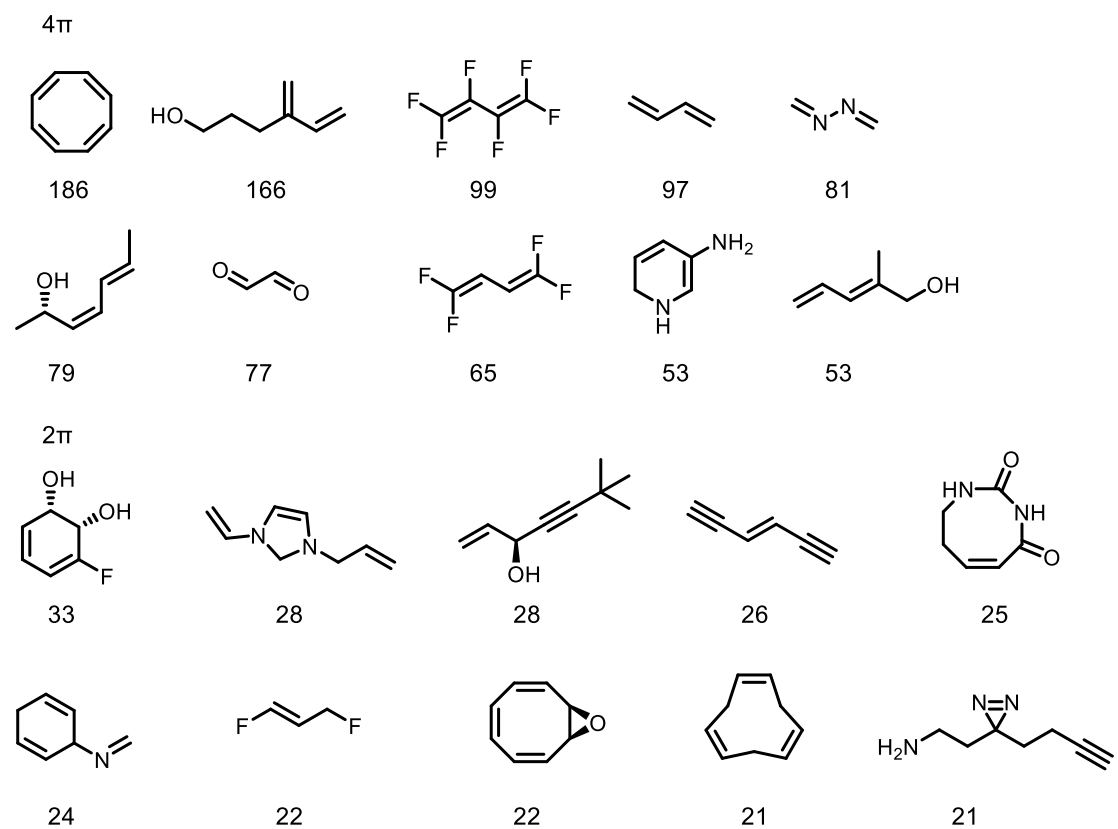


Figure SN The Most Frequently Occurring 4 π and 2 π Structures in DFT Accessible Dataset

Section 3 Details of Machine Learning Model Construction

The full reaction space comprises over 1.28 million unique reactant combinations, which, when factoring in site selectivity and stereoselectivity, expands to more than 10 million distinct Diels-Alder reactions. Performing DFT calculations to determine the reaction Gibbs free energy (ΔG_{rxn}), activation Gibbs free energy ($\Delta G^\ddagger$), distortion energy, and interaction energy for each of these reactions would demand an impractical amount of computational time and resources. To overcome this challenge, we leveraged machine learning models to predict these energy-related properties. By computing DFT data for just 10^4 carefully selected reactions, the models enabled reasonably accurate predictions across the entire reaction space, significantly reducing computational overhead.

Section 3.1 Details of High-Throughput DFT Calculations

To calculate the reaction Gibbs free energy (ΔG_{rxn}), activation Gibbs free energy ($\Delta G^\ddagger$), distortion energy, and interaction energy, the structures of reactants, products, and transition states must be fully optimized. Manually constructing and validating transition states for over 50,000 DA reactions poses a significant challenge due to the scale and complexity involved. To address this, we developed a Python-based automated workflow to generate and validate the structures of reactants, products, and transition states, ensuring both efficiency and structural accuracy throughout the process.

Section 3.1.1 Workflow of Reactant Conformation Construction

Initial conformations of the reactants were generated from their SMILES representations using the *EmbedMolecule* method in the RDKit module. However, this approach incorrectly produces cis-cyclooctene when applied to trans-cyclooctene due to limitations in stereochemical handling. To resolve this, we implemented a scripted workaround tailored to trans-cyclooctene: its structure was temporarily converted into a propylene oxide derivative with predefined geometry, processed via *EmbedMolecule* to establish the correct conformation, and then reverted to trans-cyclooctene by removing the oxygen atom. Following this correction, all reactant structures underwent a multi-step refinement process. This included force field optimization to relax initial geometries, an xTB-based conformational search to explore low-energy conformers, and subsequent DFT optimization to achieve high-precision structures and energies suitable for downstream analysis.

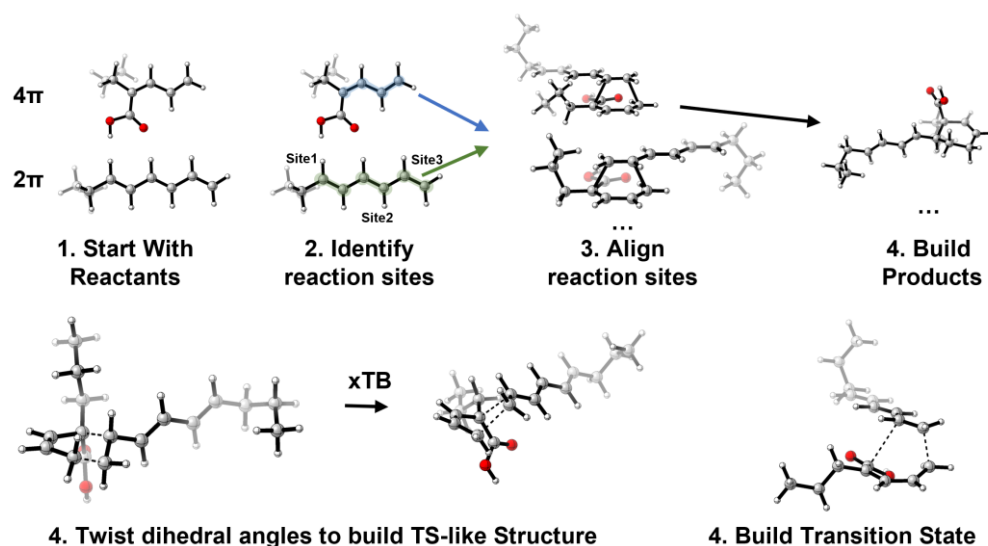
Section 3.1.2 Workflow of Product Structure Construction

The construction of product structures leverages the optimized geometries of the reactants as a starting point. Reaction sites—defined as the specific atomic positions undergoing chemical transformation—were first identified on both the 4π and 2π systems. All possible pairwise combinations of these sites were then systematically evaluated by aligning the 2π plane above the 4π plane, with the reactive atoms positioned 1.5 Å apart to simulate the initial approach. To account for stereochemical selectivity, the orientations of the 2π and 4π systems were iteratively flipped during alignment. Bonding relationships at the reaction sites were subsequently established and adjusted using RDKit's molecular manipulation tools. The resulting structures underwent force field optimization to eliminate steric clashes, followed by xTB conformational searches to efficiently sample the potential energy surface. Finally, DFT optimization was performed to produce fully relaxed product geometries with accurate energies.

Section 3.1.3 Workflow of Transition State Structure Construction

Transition state optimization begins with the optimized product structure as a reference. Using xTB

optimization, bonds in the corresponding reactive regions were elongated to mimic the partial bond formation characteristic of the transition state, while dihedral angles near the reaction sites were systematically twisted to shift chair-like conformations into boat-like conformations, aligning with the stereoelectronic requirements of DA reactions. These modified structures then underwent restricted DFT optimizations to constrain the geometry along the reaction coordinate, followed by rigorous transition state searches to locate and refine the saddle points. This multi-step process ensured accurate representation of the intermediate states along the reaction pathway, capturing the energetic and structural features of the transition states.



Section 3.1.4 Details of Force Field Optimization and xTB Conformational Search

The initial optimization phase employed molecular mechanics with the Universal Force Field (UFF) as implemented in RDKit. This step aimed to refine the geometries of reactants and products using classical force field principles, yielding structurally plausible configurations. By minimizing steric and energetic distortions, this process established a reliable starting point for subsequent quantum mechanical calculations, ensuring consistency and reducing the likelihood of convergence issues in higher-level optimizations.

The conformational search was conducted using the xTB software (version 6.4.0) in conjunction with the CREST tool (version 2.12), employing the GFN2-xTB method. This semi-empirical quantum mechanical approach enabled an efficient and systematic exploration of the potential energy surface to identify low-energy conformers. To quantify structural diversity, the root-mean-square deviation (RMSD) between generated conformers was calculated using RDKit's *Chem.AllChem.GetConformerRMS* function. A threshold RMSD of 1.5 Å was applied as a criterion to select up to three distinct conformers per molecule, balancing conformational variety with computational feasibility. These selected conformers were retained for further refinement via DFT optimization, ensuring that the subsequent quantum mechanical analysis captured a representative range of accessible geometries while minimizing redundancy.

Section 3.1.5 Details of DFT Optimization

All density functional theory (DFT) calculations were performed using the Gaussian 16 software package (Revision C.01). Geometry optimizations of energy minima (reactants and products) and transition states, as well as subsequent single-point energy calculations, were conducted consistently at a uniform level of theory to ensure comparability. The B3LYP exchange-correlation functional was employed, paired with the 6-31G(d) basis set, and augmented with the D3 dispersion correction including Becke-Johnson damping (D3BJ) to account for non-covalent van der Waals interactions. Vibrational frequency analyses were carried out at this level of theory to confirm the nature of each stationary point: energy minima were identified by the absence of imaginary frequencies, while transition states were characterized by a single imaginary frequency corresponding to the reaction coordinate. These analyses also provided thermal corrections to the Gibbs free energy at 298.15 K and 1 atm, incorporating zero-point energy, enthalpy, and entropy contributions.

To enhance the accuracy of electronic energies, single-point energy calculations were performed using the B3LYP functional with the more extensive 6-311+G(d,p) basis set, again incorporating the D3BJ dispersion correction. Solvent effects in aqueous media were modeled during these single-point calculations using the self-consistent reaction field (SCRF) approach with the SMD (Solvation Model based on Density) continuum solvation model, parameterized for water ($\epsilon = 78.3553$). This combination of methods ensured a robust treatment of both electronic and environmental effects, critical for accurately assessing reaction energetics in a biologically relevant context.

Section 3.1.6 Calculation of Distortion/Interaction Energy

The 4π and 2π distortion energies were calculated by extracting the respective 4π and 2π fragments from the optimized transition state geometry in the gas phase. Single-point energy calculations were performed on these isolated fragments at the B3LYP/6-311+G(d,p)-D3BJ level of theory, consistent with the methodology applied to the full systems. The distortion energy for each component was then determined by subtracting the single-point energy of its fully optimized ground-state geometry (obtained at the same level of theory) from the energy of the distorted fragment at the transition state. To maintain consistency in model training and data management, interaction energies were defined with a negative sign convention. These were computed as the difference between the activation energy ($\Delta E^\ddagger$, derived from electronic energies at the transition state relative to the separated reactants) and the sum of the 4π and 2π distortion energies, effectively accounting for the stabilizing electronic interactions in the transition state. While the calculated interaction energies are predominantly negative—reflecting attractive forces—their absolute (positive) values were used exclusively for visualization purposes to illustrate their distribution and magnitude effectively.

Section 3.1.7 Verification Process for Optimization Results

To ensure the structural integrity of reactants and products during optimization, we developed an automated Python script to detect unintended bond breaking or formation. The script assesses bonding relationships by comparing interatomic distances before and after optimization. For each bond in a molecule, the distance between the bonded atoms was evaluated against predefined thresholds: if the optimized distance decreased to less than 0.75 times the original bond length (indicative of spurious bond formation) or exceeded 1.6 times the original length (suggesting bond cleavage), the molecule was flagged as having undergone an unacceptable structural change and was excluded from the dataset. These thresholds were selected to account for typical vibrational stretching while identifying significant

deviations.

For transition states, the validation process mirrored that of reactants and products, with additional criteria applied to partially formed bonds critical to the Diels-Alder reaction coordinate. Excluding these partial bonds, the remaining bonding network was subjected to the same distance-based checks. For a transition state to be considered valid, the bond lengths of partially formed bonds were required to lie within 1.6 Å to 4.0 Å, reflecting the expected range for a forming σ -bond at the saddle point. Furthermore, vibrational frequency analysis was conducted to confirm a single imaginary frequency below -300 cm^{-1} , indicative of a first-order saddle point. The associated vibrational mode was required to displace the atoms involved in the partially formed bonds by at least 0.4 Å along the reaction coordinate, ensuring the mode corresponds to bond formation or breaking. Additionally, the cis-trans configurations of double bonds in the 4π and 2π components were verified to remain consistent with their initial geometries, preserving stereochemical fidelity.

In cases where the bond length range (1.6–4.0 Å) and cis-trans configurations were satisfied, but the imaginary frequency either exceeded -300 cm^{-1} or failed to alter the interatomic distance of partially formed bonds by at least 0.4 Å, further validation was performed using intrinsic reaction coordinate (IRC) analysis. The IRC was computed in both forward and reverse directions from the transition state using the B3LYP/6-31G(d)-D3BJ level of theory, consistent with the optimization protocol. If the resulting structures in either direction converged to a valid product geometry—matching the expected connectivity and energy minimum—the transition state was deemed structurally correct and retained. This multi-tiered approach ensured robust verification of transition states across the dataset.

Section 3.2 Details of All Descriptors used in model.

The predictive model integrates three distinct classes of molecular descriptors: physical organic chemistry-based descriptors, molecular fingerprint descriptors, and pre-trained deep learning molecular fingerprint descriptors. Each category offers unique perspectives on molecular properties, collectively enabling a comprehensive and multidimensional representation of chemical structures for robust reactivity prediction.

Section 3.2.1 Details and calculations of Physical Organic descriptors

A total of 65 physical organic descriptors were selected, spanning five key categories: frontier orbital energies, atomic Mulliken charges, interatomic distances, bond angles, Sterimol parameters, and dipole moments. These descriptors capture both site-specific local properties—such as atomic charges, interatomic distances, bond angles, and Sterimol steric parameters—and global molecular attributes, including frontier orbital energies and dipole moments.

To illustrate the assignment of these descriptors, Figure S1 depicts a representative Diels-Alder reaction between a 4π system and a 2π system, defining the reaction sites explicitly. For the 4π system, the reactive atoms are labeled [1, 2, 3, 4], while for the 2π system, they are [11, 12]. Adjacent atoms influencing local reactivity—such as [5, 6, 7, 8, 9, 10] in the 4π system and [13, 14, 15, 16] in the 2π system—are also included in the analysis. The descriptors described herein adhere to this numbering convention for consistency across the dataset. This systematic framework ensures that the descriptors effectively encapsulate both the localized reactivity at the reaction sites and the broader molecular context, providing

a solid basis for modeling chemical behavior and reactivity trends.

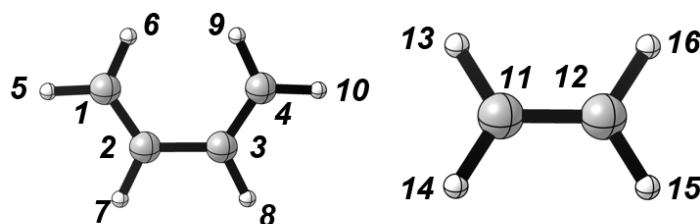


Figure S1 Typical Example Diels-Alder 4π and 2π with atomic ID

Section 3.2.1.1 Atomic Charge (Length 14)

Mulliken atomic charges were calculated for atoms in both 4π and 2π , as detailed in Table SN. These charges serve as critical descriptors for Diels-Alder (DA) reactions by quantifying the electron density distribution at and near the reaction sites, thereby reflecting the electrophilic or nucleophilic character of the 4π and 2π systems. This local electronic information enhances the model's ability to predict reactivity trends and transition state stabilization.

Table SN Details of Atomic Charge Descriptors

Descriptor Name	Detail of Descriptor
diene_charge_a	Mulliken Charge of atom 1
diene_charge_b	Mulliken Charge of atom 2
diene_charge_c	Mulliken Charge of atom 3
diene_charge_d	Mulliken Charge of atom 4
diene_max_neighbor_charge	Maximum Charge of [atom 5, atom 6, atom 7, atom 8, atom 9, atom 10]
diene_min_neighbor_charge	Minimum Charge of [atom 5, atom 6, atom 7, atom 8, atom 9, atom 10]
diene_max_charge	Maximum Charge of All Atoms in 4π
diene_min_charge	Minimum Charge of All Atoms in 4π
ene_charge_e	Mulliken Charge of atom 11
ene_charge_f	Mulliken Charge of atom 12
ene_max_neighbor_charge	Maximum Charge of [atom 13, atom 14, atom 15, atom 16]
ene_min_neighbor_charge	Minimum Charge of [atom 13, atom 14, atom 15, atom 16]
ene_max_charge	Maximum Charge of All Atoms in 2π
ene_min_charge	Minimum Charge of All Atoms in 2π

Section 3.2.1.2 Atomic Distance (Length 5)

Interatomic distances were calculated for a selected set of five atom pairs, as specified in Table SN. These distances serve as essential descriptors for Diels-Alder (DA) reactions, which reflects strain effects and proximity effects, critical for assessing reactivity and transition state feasibility in the predictive model.

Table SN Details of Atomic Distance Descriptors

Descriptor Name	Detail of Descriptor
bond1	Atomic Distance Between atom 1 and atom 2
bond2	Atomic Distance Between atom 2 and atom 3

bond3	Atomic Distance Between atom 3 and atom 4
distance_ad	Atomic Distance Between atom 1 and atom 4
bond4	Atomic Distance Between atom 11 and atom 12

Section 3.2.1.3 Bond Angle (Length 6)

Bond angles were calculated for a selected set of several atom triplets, as detailed in Table SN. These angles serve as key descriptors for Diels-Alder (DA) reactions by quantifying the angular geometry around the reaction sites in the 4π and 2π systems. Specifically, they reflect the degree of strain and conformational rigidity at these positions, providing insight into steric and electronic factors that influence reactivity and transition state accessibility in the predictive model.

Table SN Details of Bond Angle Descriptors

Descriptor Name	Detail of Descriptor
angle_a	Angle of [Atom1, Atom2, Atom3]
angle_b	Angle of [Atom2, Atom3, Atom4]
diene_max_angle	Maximum Angle of [5,1,2], [6,1,2], [9,4,3] and [10,4,3]
diene_min_angle	Minimum Angle of [5,1,2], [6,1,2], [9,4,3] and [10,4,3]
ene_max_angle	Maximum Angle of [13,11,12], [14,11,12], [16,12,11] and [15,12,11]
ene_min_angle	Minimum Angle of [13,11,12], [14,11,12], [16,12,11] and [15,12,11]

Section 3.2.1.4 Dihedral Angle (Length 1)

Dihedral angles were calculated for a selected set of atom quartets, as specified in Table SN. These angles serve as descriptors for Diels-Alder (DA) reactions by capturing the torsional geometry around reaction sites, offering insight into conformational flexibility and steric effects critical to reactivity prediction.

Table SN Details of Dihedral Angle Descriptors

Descriptor Name	Detail of Descriptor
Torsion	Dihedral Angle of [Atom1, Atom2, Atom3, Atom4]

Section 3.2.1.5 Sterimol Descriptors (Length 12)

Sterimol descriptors, specifically the substituent parameters L, B1, and B5, were calculated based on the optimized geometries of the reactants. These descriptors quantify the steric effects of substituents attached to the reaction sites of the 4π and 2π systems, providing critical insights into spatial constraints that influence Diels-Alder (DA) reactivity and transition state stability.

Table SN Details of Bond Angle Descriptors

Descriptor Name	Detail of Descriptor
max_diene_L	Maximum Value of [L (1,5), L (1,6), L (4,9), L (4,10)]
min_diene_L	Minimum Value of [L (1,5), L (1,6), L (4,9), L (4,10)]
max_diene_B1	Maximum Value of [B1 (1,5), B1 (1,6), B1 (4,9), B1 (4,10)]
min_diene_B1	Minimum Value of [B1 (1,5), B1 (1,6), B1 (4,9), B1 (4,10)]
max_diene_B5	Maximum Value of [B1 (1,5), B1 (1,6), B1 (4,9), B1 (4,10)]
min_diene_B5	Minimum Value of [B1 (1,5), B1 (1,6), B1 (4,9), B1 (4,10)]
max_ene_L	Maximum Value of [L (11,13), L (11,14), L (12,15), L (12,16)]

min_ene_L	Minimum Value of [L (11,13), L (11,14), L (12,15), L (12,16)]
max_ene_B1	Maximum Value of [B1 (11,13), B1 (11,14), B1 (12,15), B1 (12,16)]
min_ene_B1	Minimum Value of [B1 (11,13), B1 (11,14), B1 (12,15), B1 (12,16)]

Section 3.2.1.6 Orbit Energy and Dipole Moment (Length 10)

Frontier orbital energies, specifically those of the HOMO-1, HOMO, LUMO, and LUMO+1 orbitals (reported in Hartree), along with the molecular dipole moment (in Debye), were calculated for both the 4π and 2π systems. These 10 descriptors provide critical insights into the electronic structure and polarity of the reactants. The orbital energies reflect the reactivity and electronic compatibility of the 4π and 2π components in Diels-Alder (DA) reactions, while the dipole moment captures global charge distribution, influencing intermolecular interactions. All values were derived from DFT calculations at the B3LYP/6-31G(d)-D3BJ level of theory, as detailed in Section 3.1, and are summarized in Table SN.

Descriptor Name	Detail of Descriptor
diene_homo-1	The Energy of HOMO – 1 Orbit of 4π
diene_homo	The Energy of HOMO Orbit of 4π
diene_lumo	The Energy of LUMO Orbit of 4π
diene_lumo+1	The Energy of LUMO + 1 Orbit of 4π
ene_homo-1	The Energy of HOMO – 1 Orbit of 2π
ene_homo	The Energy of HOMO Orbit of 2π
ene_lumo	The Energy of LUMO Orbit of 2π
ene_lumo+1	The Energy of LUMO + 1 Orbit of 2π
diene_dipole	The Dipole Moment of 4π
ene_dipole	The Dipole Moment of 2π

Section 3.2.1.7 SVO Descriptors(Length 16)

The StereoVolume Occupancy (SVO) descriptor was developed to quantify steric hindrance across different conformations of reactants, enabling accurate predictions of reaction barrier variations, particularly those arising from conformational differences such as exo versus endo configurations in Diels-Alder (DA) reactions. By capturing the spatial distribution of molecular volume around reactive sites in three-dimensional space, the SVO descriptor provides a nuanced representation of steric effects that influence transition state energetics. The descriptor's core methodology relies on a grid-based integration approach, which calculates the buried volume within eight sub-cubes surrounding the reaction sites, yielding a detailed steric profile.

The SVO descriptor is computed in two key steps: structure alignment and grid-based integration.

Structure Alignment: The reactive sites of each reactant are first identified—specifically, atoms [1, 4] for the 4π system and atoms [11, 12] for the 2π system, consistent with the numbering in Figure S1. To standardize their orientation, the molecules undergo rotation and translation. For the 4π system, atoms 1 and 4 are aligned along the x-axis, symmetrically centered at the origin (0, 0, 0), with atom 2 positioned in the xy-plane along the negative y-axis. This orientation positions the 2π system to approach from the positive z-axis. Conversely, for the 2π system, atoms 11 and 12 are aligned along the x-axis, centered at the origin, with atom 13 placed in the xy-plane along the positive y-axis, allowing the 4π system to

approach from the negative z-axis. This consistent alignment ensures a uniform reference frame for evaluating steric interactions between the 4π and 2π partners.

Grid-Based Integration: Following alignment, a cubic region of $6 \text{ \AA} \times 6 \text{ \AA} \times 6 \text{ \AA}$ is defined, centered on the reactive sites, and discretized into a grid of 27,000 points ($30 \times 30 \times 30$) for numerical integration. This cube is subdivided into eight smaller sub-cubes, each with an edge length of $3 \text{ \AA}$, spanning the octants around the origin. The molecular volume within each sub-cube is calculated using the van der Waals radii of the atoms, providing a localized measure of steric occupancy. This process generates a 16-dimensional SVO descriptor—two sets of eight values for the 4π and 2π systems—effectively encoding the three-dimensional steric environment and its impact on reaction kinetics.

The resulting 16 SVO descriptors enhance the model's ability to predict reaction barriers by capturing conformational variability and steric hindrance with high spatial resolution. This approach is particularly valuable for distinguishing subtle steric influences on DA reactivity, such as those between exo and endo transition states, thereby improving the robustness of the predictive framework.

Section 3.2.1.8 Model-Predicted ΔG_{rxn} (Length 1)

Drawing on the Hammond postulate, which suggests a correlation between the activation energy barrier ($\Delta G^\ddagger$) and the reaction free energy change (ΔG_{rxn}), we developed a machine learning model to predict ΔG_{rxn} as a foundational step to enhance the accuracy of $\Delta G^\ddagger$ predictions. The predicted ΔG_{rxn} values were subsequently integrated as a key descriptor into the model for $\Delta G^\ddagger$, leveraging this thermodynamic relationship to improve overall predictive performance.

To enrich the training dataset, we expanded it by including 41,887 reactions with ΔG_{rxn} values exceeding 10 kcal/mol, which had been excluded from prior transition state calculations due to their thermodynamic inaccessibility. This augmentation broadened the chemical space covered by the dataset, strengthening the model's ability to predict ΔG_{rxn} accurately and enabling effective screening of bioorthogonal reactions by filtering out those with prohibitively high ΔG_{rxn} values.

For model development, k-fold cross-validation was performed using a combined dataset of 41,887 high- ΔG_{rxn} reactions and 8,273 reactions from the original training set. Physical organic (PhysOrg) descriptors, as detailed in Sections 3.2.1.1–3.2.1.7, served as input features. The resulting machine learning model generated ΔG_{rxn} predictions for both the training and test sets. These predicted values were appended to the PhysOrg descriptors of each dataset, refining the subsequent prediction of $\Delta G^\ddagger$ by providing an additional thermodynamic constraint.

For large-scale predictions across the full reaction space, ΔG_{rxn} values for uncomputed reactions were estimated using the same PhysOrg descriptors (Sections 3.2.1.1–3.2.1.7). These predicted ΔG_{rxn} values were then incorporated into the reactant descriptor set to support $\Delta G^\ddagger$ prediction. The inclusion of the 41,887 additional reactions significantly improved the model's out-of-sample performance, enhancing the accuracy of predictions for ΔG_{rxn} , $\Delta G^\ddagger$, and distortion/interaction (D/I) energies, thereby ensuring robust applicability across diverse reaction profiles.

Section 3.2.2 Correlation Analysis of PhysOrg Descriptors

To evaluate the interdependence of the physical organic (PhysOrg) descriptors (detailed in Sections 3.2.1.1–3.2.1.7), pairwise correlation coefficients were calculated using the corr function from the pandas library, applied to a dataset of 8,273 reactions. The analysis revealed that the highest correlation coefficient among descriptor pairs was 0.91, well below the threshold typically required to ensure descriptor independence in machine learning models. This modest level of correlation indicates that the PhysOrg descriptors encapsulate diverse and complementary aspects of molecular properties, reducing the risk of multicollinearity and enhancing the robustness and predictive power of the model.

A heatmap, presented in Figure S2, visually depicts the pairwise correlations among all PhysOrg descriptors. The color intensity reflects the absolute magnitude of the correlation coefficients, with darker shades corresponding to stronger correlations. As shown, most descriptor pairs exhibit weak to moderate correlations, with few approaching the maximum value of 0.91. This distribution underscores the suitability of these descriptors as independent and informative features for reliable model training and reactivity prediction.

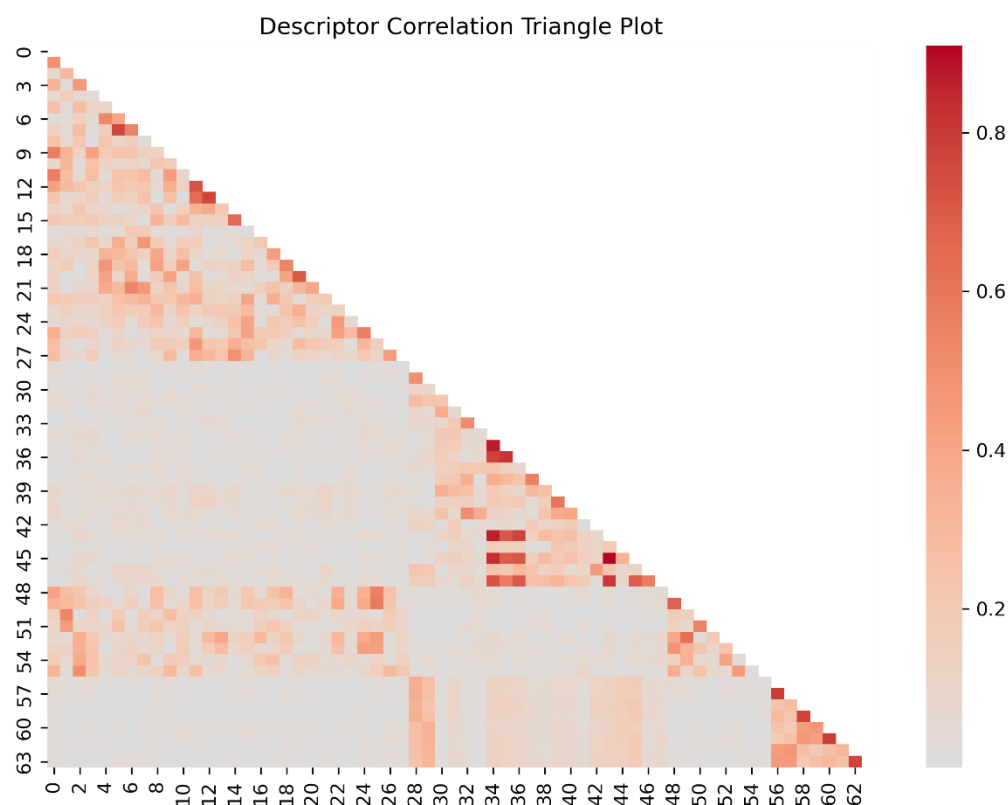


Figure S2 Heatmap of Descriptor Correlations

Section 3.2.3 Details of Other Descriptors

To establish baseline descriptors, a range of widely adopted molecular fingerprints was evaluated, encompassing 2D descriptors, 3D descriptors, and hybrid combinations of both 2D and 3D features. For each reaction, the molecular fingerprints of the 4π system, 2π system, and corresponding product were concatenated to form a unified descriptor set. The specific molecular fingerprints tested, along with their generation packages and associated parameters, are detailed in Table S1. To reduce dimensionality and facilitate analysis, all molecular fingerprint descriptors described in Section 3.2.3 were subjected to principal component analysis (PCA), decomposing them into 300 principal components.

Section 3.2.3.1 Molecular Fingerprints Descriptors

Table S1 Baseline Molecular Fingerprints Descriptors

Descriptor Name	Package	Parameters	Original Length
RDKit FP (2D)	rdkit	size=2048	2048 * 3
MorganFP (2D)	rdkit	radius=2,nBits=2048,useChirality=True	2048 * 3
RDKit Descriptor (2D)	rdkit	--	208 * 3
Modred Descriptor(2D)	mordred	ignore_3D=True	1613 * 3
Atom-centered Symmetry Functions (ACSF, 3D)	DScribe	rcut = 6.0, g2_params = [[1, 1], [1, 2], [1, 3]], g4_params = [[1, 1, 1], [1, 2, 1], [1, 1, -1], [1, 2, -1]]	80 * 3
Smooth Overlap of Atomic Positions (SOAP, 3D)	DScribe	rcut = 6.0, nmax = 4, lmax = 3	840 * 3
Local Many-body Tensor Representation (LMBTR, 3D)	DScribe	k2 = {"geometry": {"function": "inverse_distance", "grid": {"min": 0, "max": 1, "n": 10, "sigma": 0.1}, "weighting": {"function": "exponential", "scale": 0.5, "cutoff": 1e-3}} k3 = {"geometry": {"function": "cosine", "grid": {"min": -1, "max": 1, "n": 10, "sigma": 0.1}, "weighting": {"function": "exponential", "scale": 0.5, "cutoff": 1e-3}}	570 * 3
Many-body Tensor Representation (MBTR, 3D)	DScribe	k1 = {"geometry": {"function": "atomic_number", "grid": {"min": 0, "max": 8, "n": 10, "sigma": 0.1}} k2 = {"geometry": {"function": "inverse_distance", "grid": {"min": 0, "max": 4, "n": 10, "sigma": 0.1}, "weighting": {"function": "exponential", "scale": 0.5, "cutoff": 1e-3}} k3 = {"geometry": {"function": "cosine", "grid": {"min": -1, "max": 4, "n": 10, "sigma": 0.1}, "weighting": {"function": "exponential", "scale": 0.5, "cutoff": 1e-3}}	

Details of Pre-trained Deep Learning Molecular Fingerprint Descriptors.

Code is available on https://github.com/NetPharMedGroup/publication_fingerprint.

Table S2 Pre-trained Deep Learning Molecular Fingerprint Descriptors

Descriptor Name	Model	Original Length
VAE	VAE	256 * 3
TRFM	Transformer	1024 * 3
GAE	GAE	64

Section 3.3. Details of Machine Learning Model

Section 3.3.1 Details of Traditional Machine Learning Model

To assess the performance of candidate machine learning algorithms, we employed 5-fold cross-validation with a fixed random seed of 0 to ensure reproducibility. Model performance was quantified using two regression metrics: the mean absolute error (MAE), sourced from

sklearn.metrics.mean_absolute_error, and the coefficient of determination (R^2), sourced from *sklearn.metrics.r2_score*. These metrics provided a robust evaluation of predictive accuracy and goodness-of-fit for each algorithm.

The candidate algorithms comprised 11 widely used regression models implemented in Python: Categorical Boosting Regressor (CatBoost, CB), Gradient Boosting Regressor (GB), Random Forest Regressor (RF), Adaptive Boosting Regressor (AdaBoost, ADB), Extremely Randomized Trees Regressor (Extra Trees, ET), Extreme Gradient Boosting Regressor (XGBoost, XGB), Least Absolute Shrinkage and Selection Operator Regressor (Lasso), Ridge Regression (Ridge), Bayesian Ridge Regression (BR), Support Vector Regressor (SVR), and K-Nearest Neighbors Regressor (KNN). All models were sourced from the scikit-learn library, except for CB (from the catboost package) and XGB (from the xgboost package). Hyperparameter optimization was performed using the *hyperopt* module, with parameter spaces defined for each model. The default hyperparameter settings and their corresponding ranges are detailed in Table SN.

Table SN Details of machine learning models

Model name	Details of Model Function	Hyperparameters Space
CB	CatBoostRegressor	iterations: [10000] learning_rate: [0.01] depth: [6]
GB	GradientBoostingRegressor	n_estimators: [100,200,300] min_samples_split: [2,3,4,5,6] min_samples_leaf: [1,2,3,4]
RF	RandomForestRegressor	n_estimators: [100,200,300] min_samples_split: [2,3,4,5,6] min_samples_leaf: [1,2,3,4]
ADB	AdaBoostRegressor	n_estimators: [100,200,300]
ET	ExtraTreesRegressor	n_estimators: [100,200,300] min_samples_split: [2,3,4,5,6] min_samples_leaf: [1,2,3,4]
XGB	XGBRegressor	n_estimators: [100,200,300] max_depth: [2,3,4,5,6]
Lasso	Lasso	alpha: [0,0.5,1]
Ridge	Ridge	alpha: [0,0.5,1]
BR	BayesianRidge	alpha_1: [1e-7, 1e-6, 1e-5] alpha_2: [1e-7, 1e-6, 1e-5] lambda_1: [1e-7, 1e-6, 1e-5] lambda_2: [1e-7, 1e-6, 1e-5]
SVR	SVR	C: [0, 0.5, 1] epsilon: [0.1,0.2,0.3]
KNN	KNeighborsRegressor	n_neighbors: [1,2,3,4,5,6,7,8,9,10]

And the details of the 5-Fold regression performances are shown as Figure SN~SN.

Section 3.3.2 Details and Performance of Deep Learning Model

Section 3.3.2.1 Details of AttentiveFP

The AttentiveFP model, a graph neural network designed for molecular property prediction, is accessible at <https://github.com/OpenDrugAI/AttentiveFP>. A Python script for executing AttentiveFP is provided in our GitHub repository at *Data/AttentiveFP/Run_AttentiveFP.ipynb*. The model was trained using SMILES representations of the product structures as input. Due to the computationally intensive training process, we opted for a random 8:1:1 split of the dataset into training, validation, and test sets, respectively, rather than the 5-fold cross-validation employed for other machine learning models in this study. This adjustment balanced computational efficiency with reliable performance evaluation.

Section 3.3.2.2 Details of QM-GNN

The Quantum Mechanics-Graph Neural Network (QM-GNN) model, developed by the Coley research group for predicting activation and reaction energies of [3+2] bioorthogonal reactions, is available at https://github.com/tstuyver/multitask_QM_GNN. We downloaded this model and created a custom execution script, available in our GitHub repository at *Data/multitask_QM_GNN-develop/run_GNN.ipynb*. The model utilizes atomic descriptors generated by the Python module qmdesc as input. Given the mechanistic differences between Diels-Alder (DA) and [3+2] reactions, we excluded descriptors related to ground-state and excited-state free energy differences of the reactants, tailoring the input to suit the DA reaction context while leveraging the model's predictive framework.

Section 4 Details of Model Performance

Section 4.1 Model Performance of Traditional Machine Learning Model



Figure SN R² Result of 4π Distortion Prediction

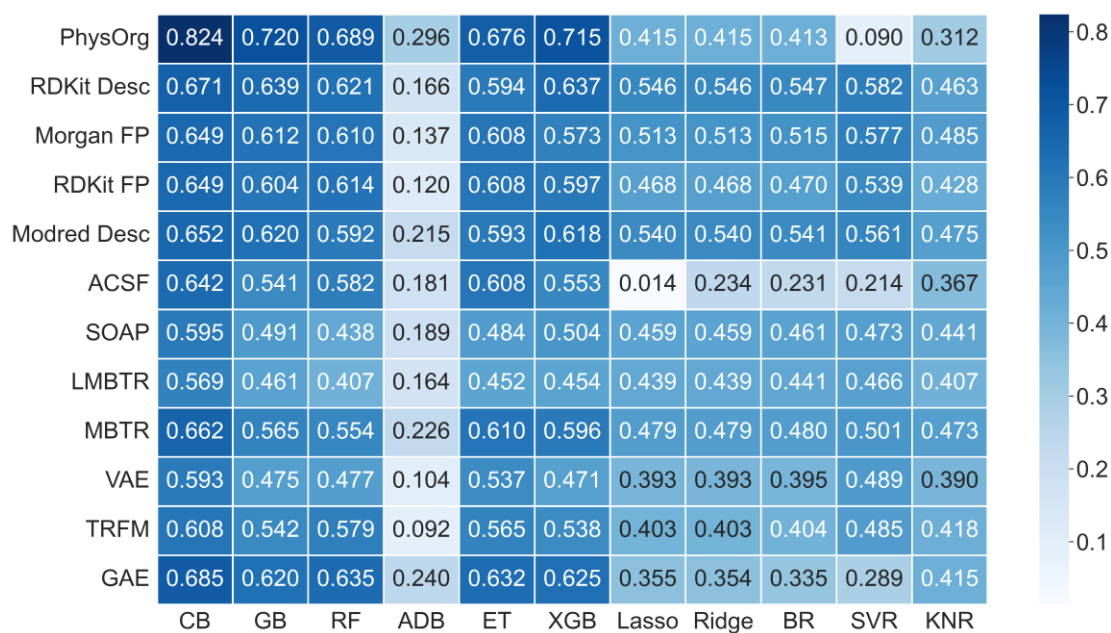


Figure SN R² Result of 2 π Distortion Prediction

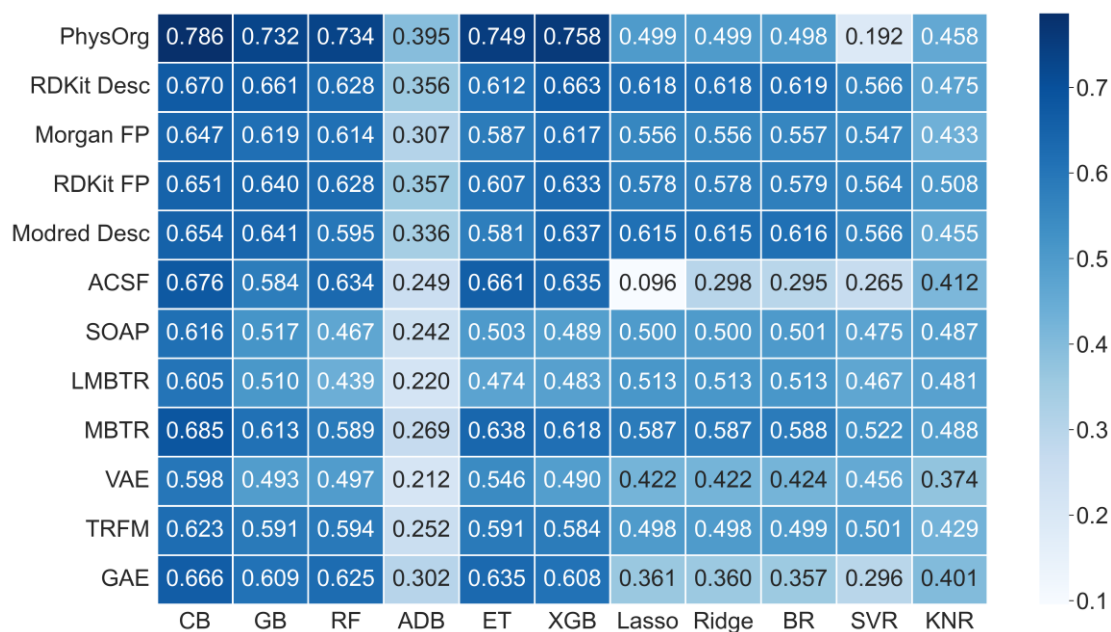


Figure SN R² Result of Interaction Prediction



Figure SN R² Result of $\Delta G^\ddagger$ Prediction

Section 4.2 Performance of AttentiveFP, QM-GNN and Our Model

For comparison, we also included the performance of a CatBoost model combined with the PhysOrg descriptors (exclude predicted ΔG_{rxn} in fair) using the same data splitting method. The results of all models are showed in Figure SN

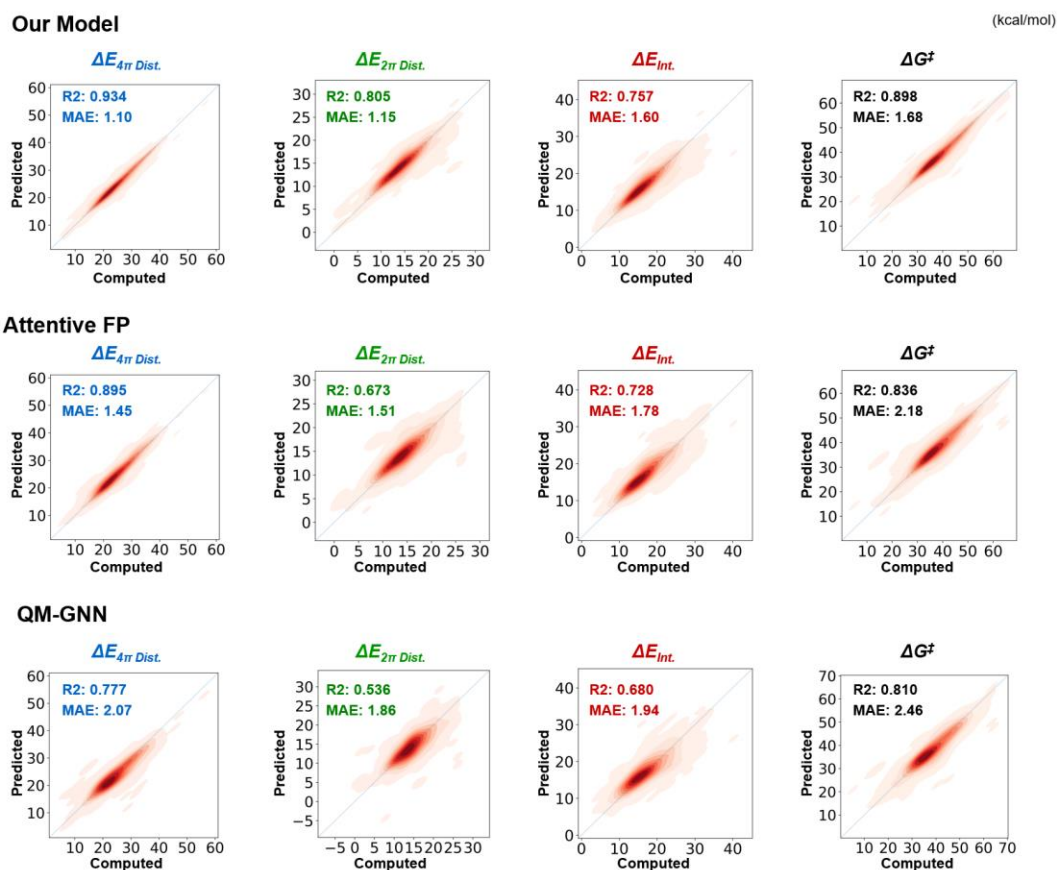


Figure SN Model Performance of AttentiveFP, QM-GNN and Our Model

Section 4.3 ΔG_{rxn} Prediction Performance

For the thermodynamic predictions, the model achieved excellent fitting accuracy in both 5-Fold cross-validation and Out-of-Sample (OOS) testing. The 5-Fold cross-validation was implemented using the K-Fold module from scikit-learn, with the random seed set to 0. In the OOS tests, 10% of the 4π and 2π datasets were randomly selected using a random seed of 0, ensuring that the test set consisted solely of these 4π and 2π samples, which were not present in the training set. All 5-Fold and OOS tests mentioned in this study were conducted using similar methods.

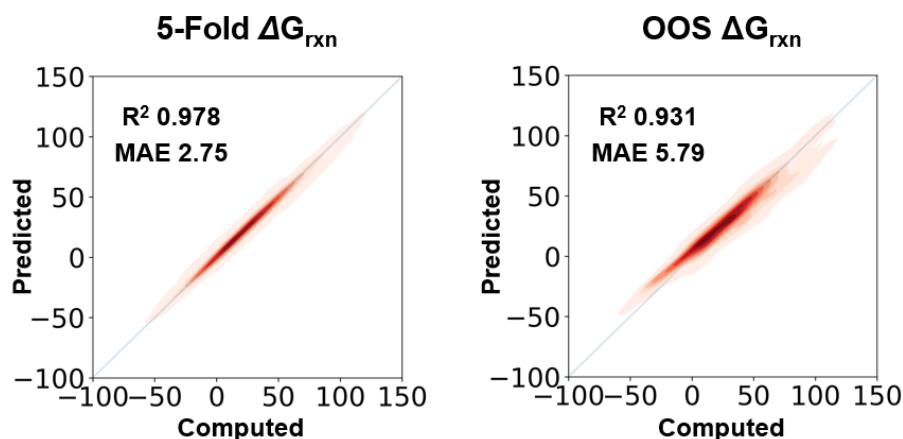


Figure SN ΔG_{rxn} Prediction in 5-Fold and OOS

Section 4.3.2 OOS Prediction Performance

The Out of Sample (OOS) prediction poses a significant challenge in both D/I and energy barrier predictions. The difficulty arises from the limited availability of transition state data combined with the diversity of reactants. However, with the assistance of thermodynamic energy predictions, the model's OOS performance improved substantially compared to previous results. This improvement was particularly notable due to the contribution of over 40,000 "discarded" thermodynamic data points, which significantly enhanced the accuracy of thermodynamic descriptors in the OOS predictions, making the model's OOS performance reliable for practical use.

8273 OOS R^2	$\Delta E_{4\pi \text{ Dist.}}$	$\Delta E_{2\pi \text{ Dist.}}$	$\Delta E_{\text{Int.}}$	$\Delta G^\ddagger$	
PhysOrg (No ΔG_{rxn})	0.582	0.606	0.641	0.695	0.8 0.7 0.6
PhysOrg (8273 ΔG_{rxn})	0.553	0.627	0.645	0.695	
PhysOrg (50164 ΔG_{rxn})	0.705	0.682	0.645	0.803	

Figure SN $\Delta G^\ddagger$ and D/I Energy Prediction in OOS

Although the model has certain limitations in Out of Sample (OOS) predictions, this did not significantly impact the prediction of D/I and energy barriers in the 1.28 million reaction space. This is because the initial selection of 5,000 structurally diverse Diels Alder (DA) reaction combinations had already explored all 718 types of 4π and 1,789 types of 2π species in the space. While the final dataset of 8,273 reactions only covered 513 stable 4π and 868 2π species as products, this was because the

remaining reactants were thermodynamically unsuitable for DA reactions. These unstable reactants were subsequently filtered out using a highly accurate ΔG_{rxn} prediction model. Therefore, the process of predicting D/I and energy barriers in the 1.28 million reaction space is not a traditional OOS process.

Section 5: Selection of Potential Bioorthogonal Reactions

We first trained the models using all available data. Specifically, we trained the ΔG_{rxn} prediction model using a total of 50,161 thermodynamic data points and trained the 4π Distortion, 2π Distortion, Interaction, and $\Delta G^\ddagger$ prediction models using 8,273 reactions. Over the entire dataset comprising all possible reactant combinations ($718\ 4\pi$ compounds $\times$ $1,789\ 2\pi$ compounds = 1,284,502 potential pairs), we used the previously trained models to predict the 4π Distortion, 2π Distortion, Interaction energy, ΔG_{rxn} , and $\Delta G^\ddagger$ for all reactions in this space. To identify reactants prone to forming dimers through Diels-Alder (DA) reactions with themselves (e.g., cyclopentadiene), we analyzed the 2507 reactants (comprising 718 4π and 1,789 2π compounds). Among these, we specifically focused on those compounds that possess a 4π structure. For these compounds, we predicted their ΔG_{rxn} and $\Delta G^\ddagger$ for DA reactions with themselves, allowing us to assess their likelihood of forming dimers. The screening process for identifying potential bioorthogonal reactants involved the following steps.

Section 5.1: Selecting Reactions with Low $\Delta G^\ddagger$ and ΔG_{rxn}

Bioorthogonal reactions must satisfy both high activity and high stability requirements. Therefore, we retained only those reactions with a $\Delta G^\ddagger$ below 24 kcal/mol and a ΔG_{rxn} below -5 kcal/mol. Specifically, for the vast number of reactions involving one 4π compound and one 2π compound that exhibit multiple selectivities, we retained only the reaction with the lowest $\Delta G^\ddagger$. Ultimately, 6,216 reactions were retained, involving 566 distinct 4π compounds and 1099 distinct 2π compounds. Results are stored in the directory “Data/Predict_DI/deltaGa_LowThan_24.csv”.

Section 5.2: Removing Reactants Prone to Self-Dimerization via DA Reactions

Some highly reactive reactants, such as cyclopentadiene, are prone to forming dimers by undergoing DA reactions with themselves. This not only makes these compounds difficult to stabilize but also introduces competition between their bioorthogonal reaction and their self-dimerization during the desired process. For a reaction to be considered bioorthogonal, it must occur efficiently without interference from side reactions like dimer formation. Therefore, any reactant capable of forming dimers via DA reactions was removed if at least one predicted $\Delta G^\ddagger$ for its self-dimerization was below 24 kcal/mol (the upper limit for bioorthogonal reaction barriers). During this process, 6 such reactants were removed from consideration.

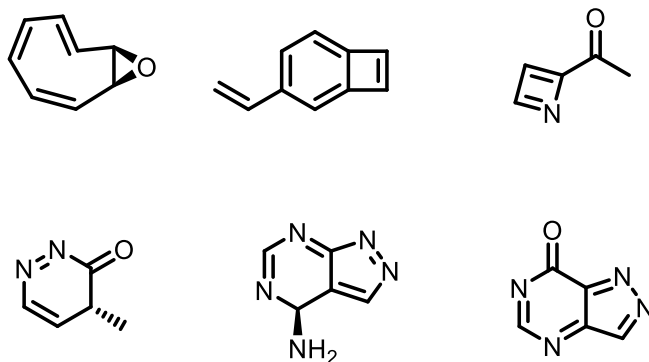


Figure SN Removed Reactants Prone to Self-Dimerization

Section 5.3: Removing Reactants Containing Specific Reactive Groups at Reaction Sites

Certain functional groups, such as carbonyls, are prone to nucleophilic attack. If a reaction site is susceptible to attack by nucleophiles present in biological systems (e.g., water or thiols), it can disrupt the bioorthogonal reaction. This issue is particularly evident for 2π compounds lacking conjugated groups that provide stability. For 2π compounds, if the reactive site consists of a non-aromatic double bond and is either C=N, C=O, N=O, N=N, or O=O, or if it is a C=C double bond conjugated with an electron-withdrawing group (e.g., C=N or C=O), such compounds are considered prone to reacting with nucleophiles and are therefore excluded. For 4π compounds, if the reactive site consists of two non-aromatic double bonds and either one of these double bonds satisfies the aforementioned criteria, the compound is deemed susceptible to nucleophilic attack and is also excluded. The aromaticity of the compounds was determined using RDKit. Ultimately, only 1071 reaction combinations were retained, involving 356 distinct 4π compounds and 46 distinct 2π compounds. Results are stored in the directory “Data/Predict_DI/deltaGa_LowThan_24_stable.csv”.

Section 5.4: Removing Other Unstable Molecules Manually

We manually inspected the remaining 356 4π and 46 2π compounds and identified several molecules with unstable structural features, such as allenes within six-membered rings. Additionally, some compounds contained N-O-N motifs, which are highly susceptible to nucleophilic attack under biological conditions. These five compounds, despite being catalogued as “In-stock” in the ZINC database, were not found in Reaxys (www.reaxys.com) with regard to their synthesis, preparation, or usage reports. This discrepancy suggests that they may be practically “unavailable” for commercial purchase. The issue likely arises from classification errors or SMILES notation inaccuracies within the ZINC database. Therefore, these compounds were removed from our dataset. Ultimately, only 429 reaction combinations were retained, involving 188 distinct 4π compounds and 41 distinct 2π compounds.

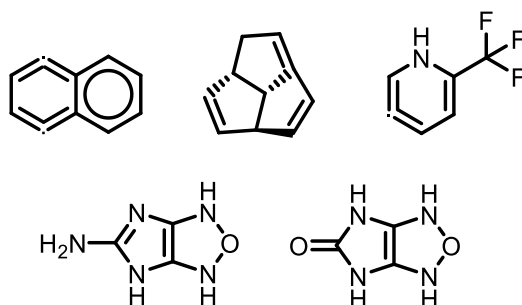


Figure S5.4 Removed Unstable Molecules

Section 5.5 Identifying Potential Bioorthogonal Reactions Based on Distortion and Interaction Energy Distributions

Building upon the high-activity bioorthogonal reactions identified earlier, we analyzed the distribution patterns of distortion and interaction energies in these reactions. By comparing these patterns with those reported in literature for known bioorthogonal molecules, we aimed to uncover unique characteristics that define bioorthogonal reactivity.

Section 5.5.1 Bioorthogonal Molecules Reported in Literature

We compiled a comprehensive list of bioorthogonal reactions from Coelho's work and other recent

studies, focusing on key bioorthogonal reactants. To simplify the analysis, we selected reactants with no more than 10 atoms as core skeletons. Using our previously trained models, we predicted the 4π Distortion, 2π Distortion, Interaction energy, ΔG_{rxn} , and $\Delta G_{\ddagger}^{\dagger}$ for all possible combinations of these bioorthogonal 4π and 2π compounds. Following the same screening criteria ($\Delta G_{\ddagger}^{\dagger} < 24$ kcal/mol and $\Delta G_{\text{rxn}} < -5$ kcal/mol), we narrowed down to 96 validated bioorthogonal reactions involving 7 distinct 4π compounds and 14 distinct 2π compounds. These selected core reactants are listed in Figure SN. Results are stored in the directory “Data/Predict_DI/ BO_LowThan_24_reported.csv”.

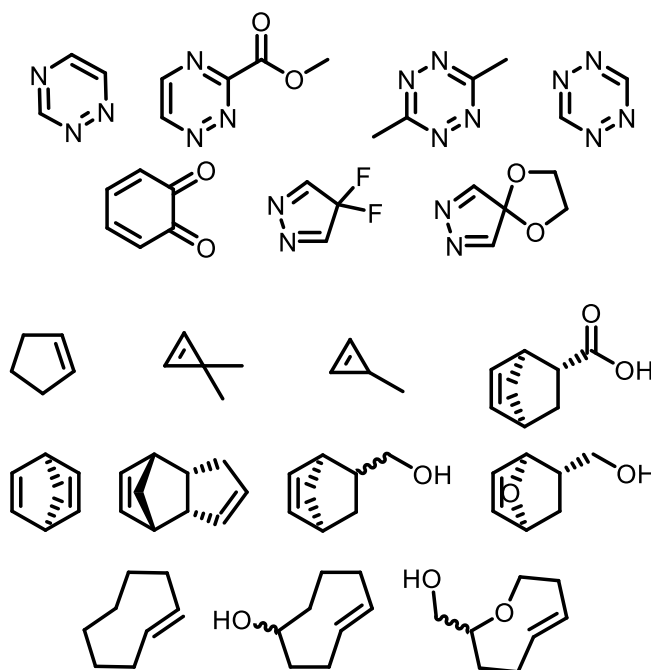
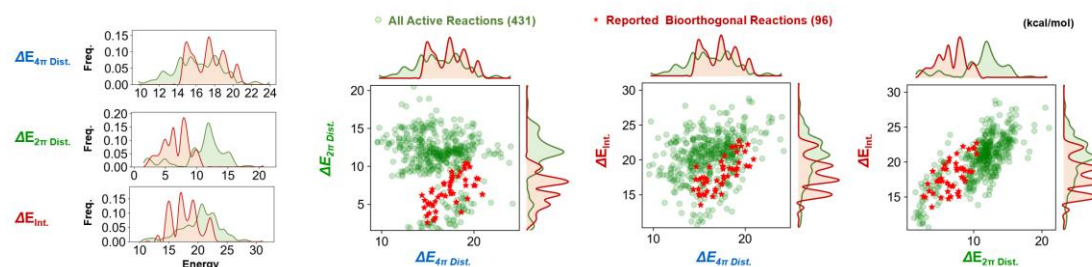


Figure SN Selected Bioorthogonal Reactants Reported in Literature

Section 5.5.2 Comparing D/I Energy Distributions Between High-Activity DA Reactions and Reported Bioorthogonal Reactions

From the initial pool of 463 high-activity bioorthogonal DA reactions, we excluded those already reported in literature, leaving us with 431 novel candidates. We analyzed the distribution patterns of distortion (D) and interaction (I) energies for these 431 reactions and compared them to the 96 validated bioorthogonal reactions identified earlier.



As shown in Figure SN, there is a noticeable difference between the D/I energy distributions of reported bioorthogonal reactions and those of high-activity DA reactions. Specifically, the reported bioorthogonal reactions tend to exhibit lower 2π Distortion values while displaying higher 4π Distortion values compared to other high-activity reactions.

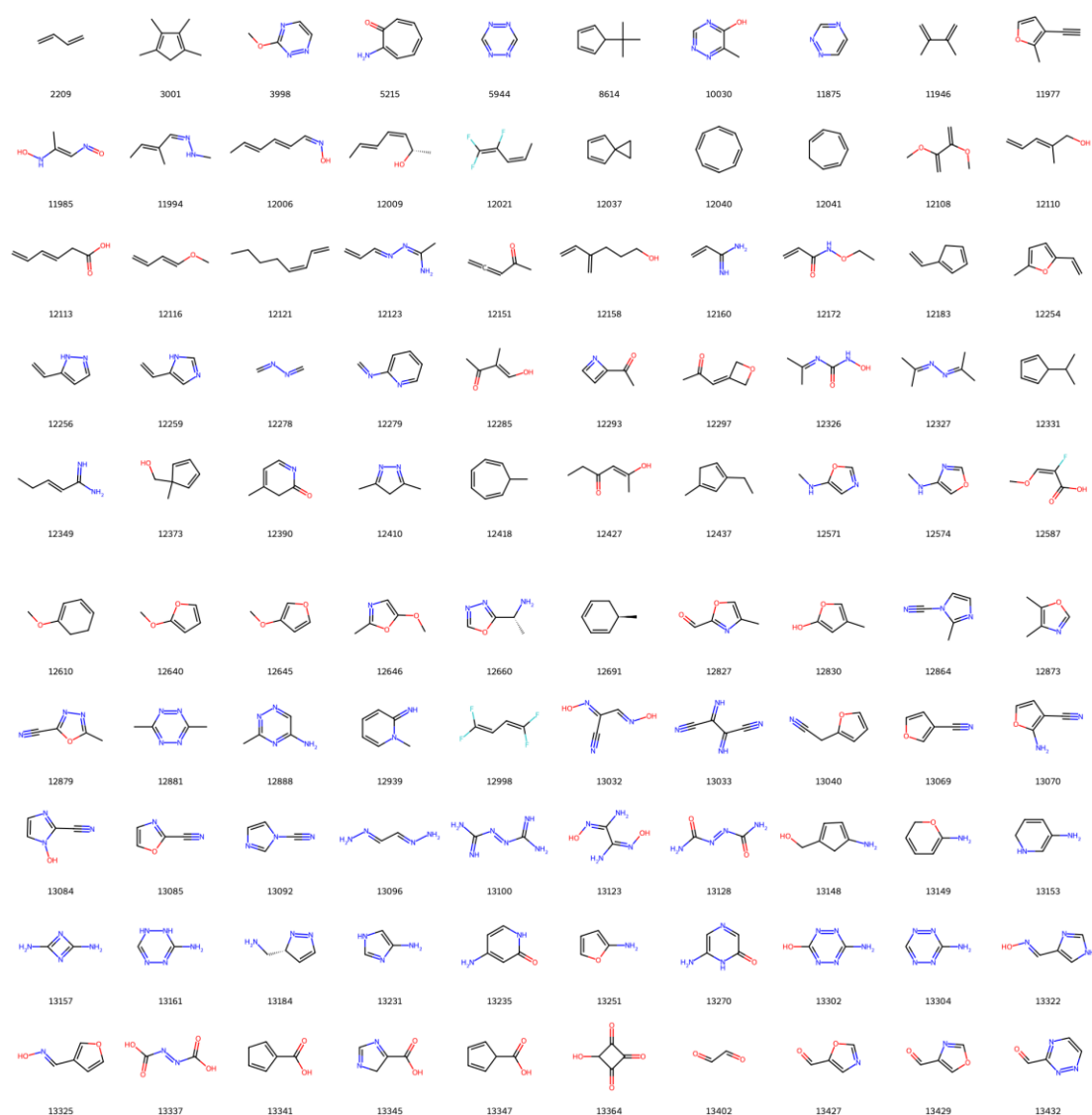
Section 5.5.3 Identifying Potential Bioorthogonal Reactants

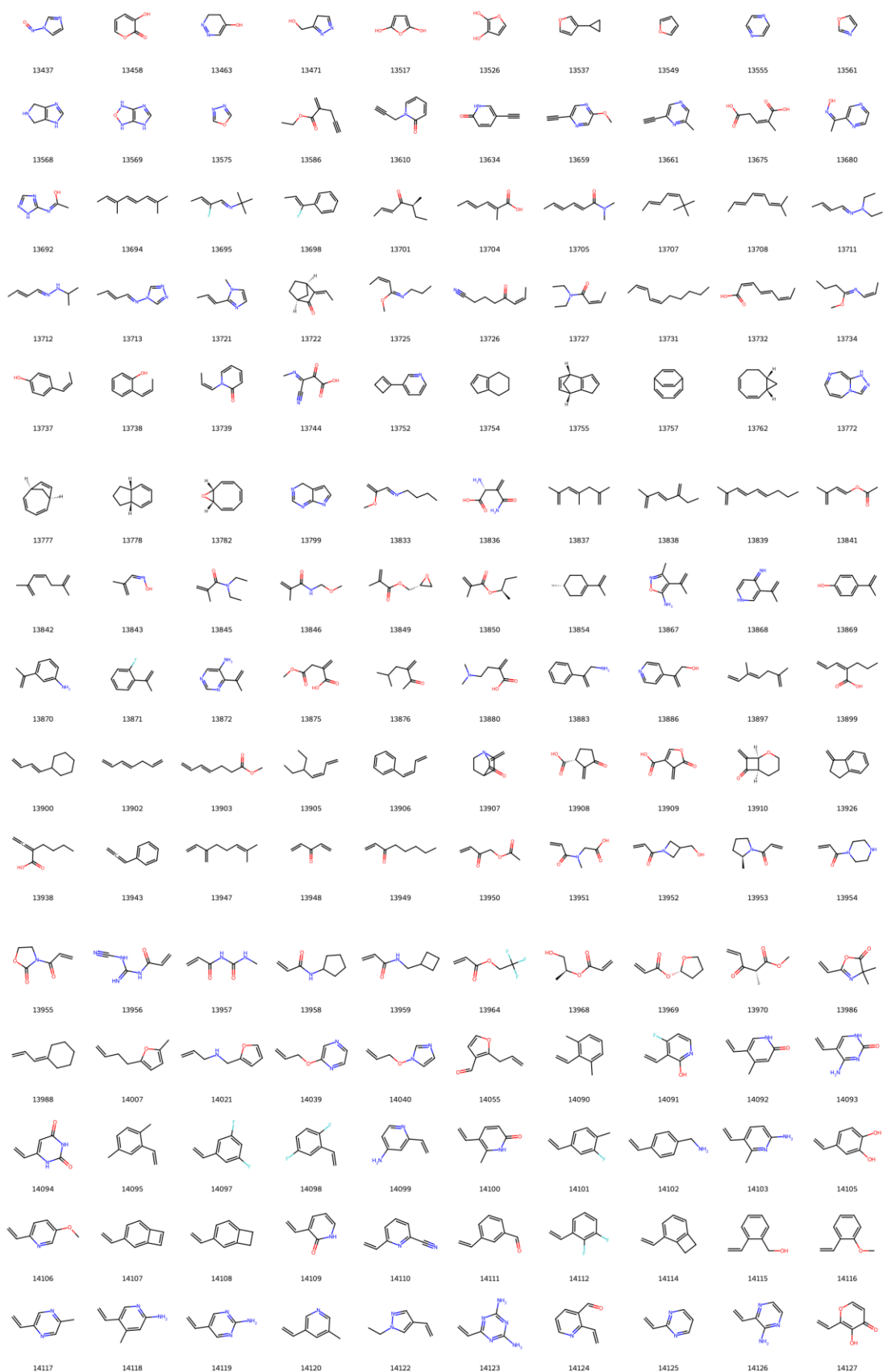
Based on the D/I energy distributions shown in Figure SN, we determined that bioorthogonal reactions should exhibit a 4π Distortion energy greater than 14 kcal/mol and a 2π Distortion energy less than 11 kcal/mol. From the previous pool of 431 reactions, we further screened to identify 126 potential bioorthogonal reactions. These reactions are composed of 85 distinct 4π compounds and 30 distinct 2π compounds.

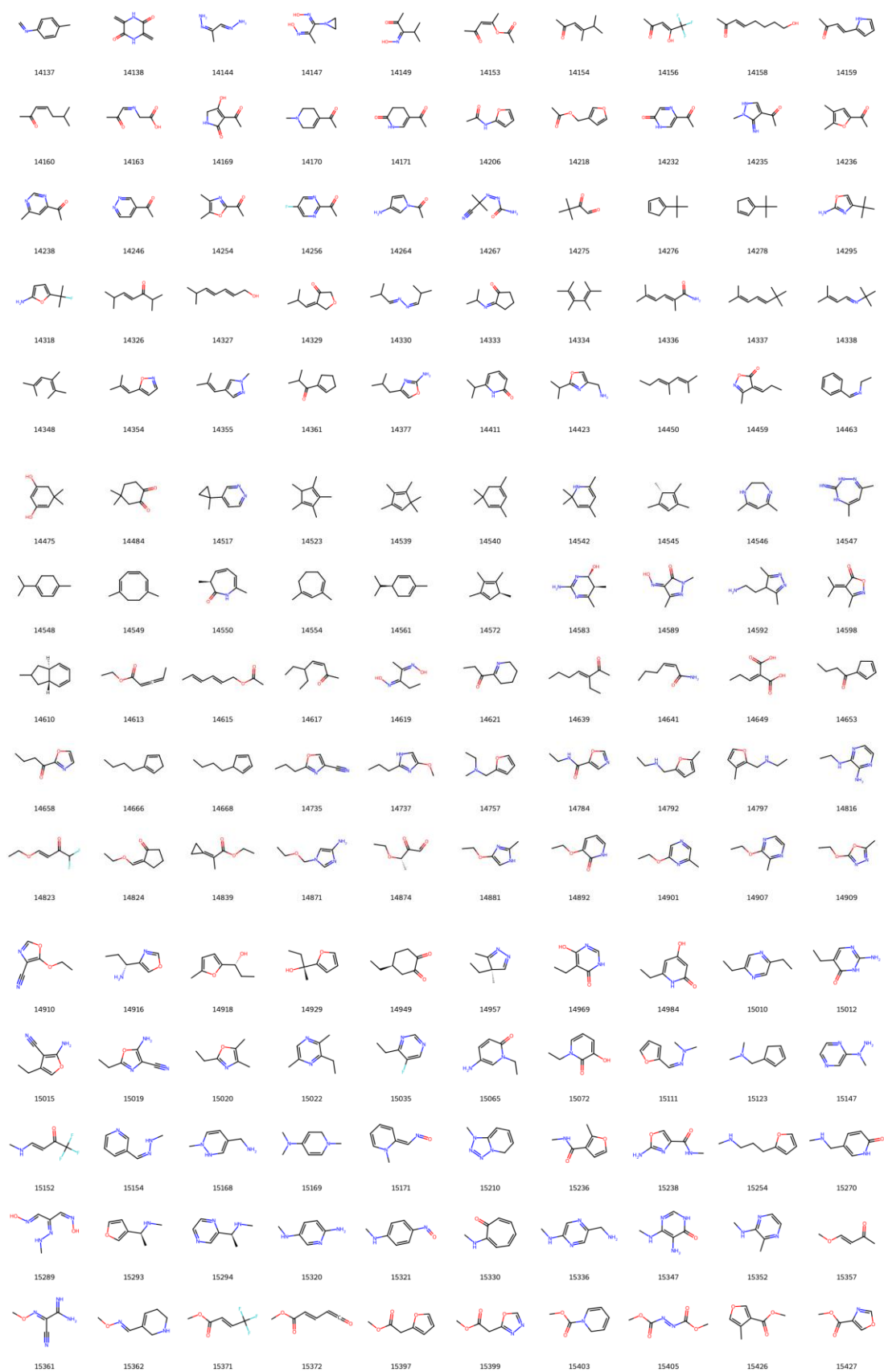
Finally, based on commercial availability, we selected specific reactions for experimental validation as our final candidates for potential bioorthogonal reactions.

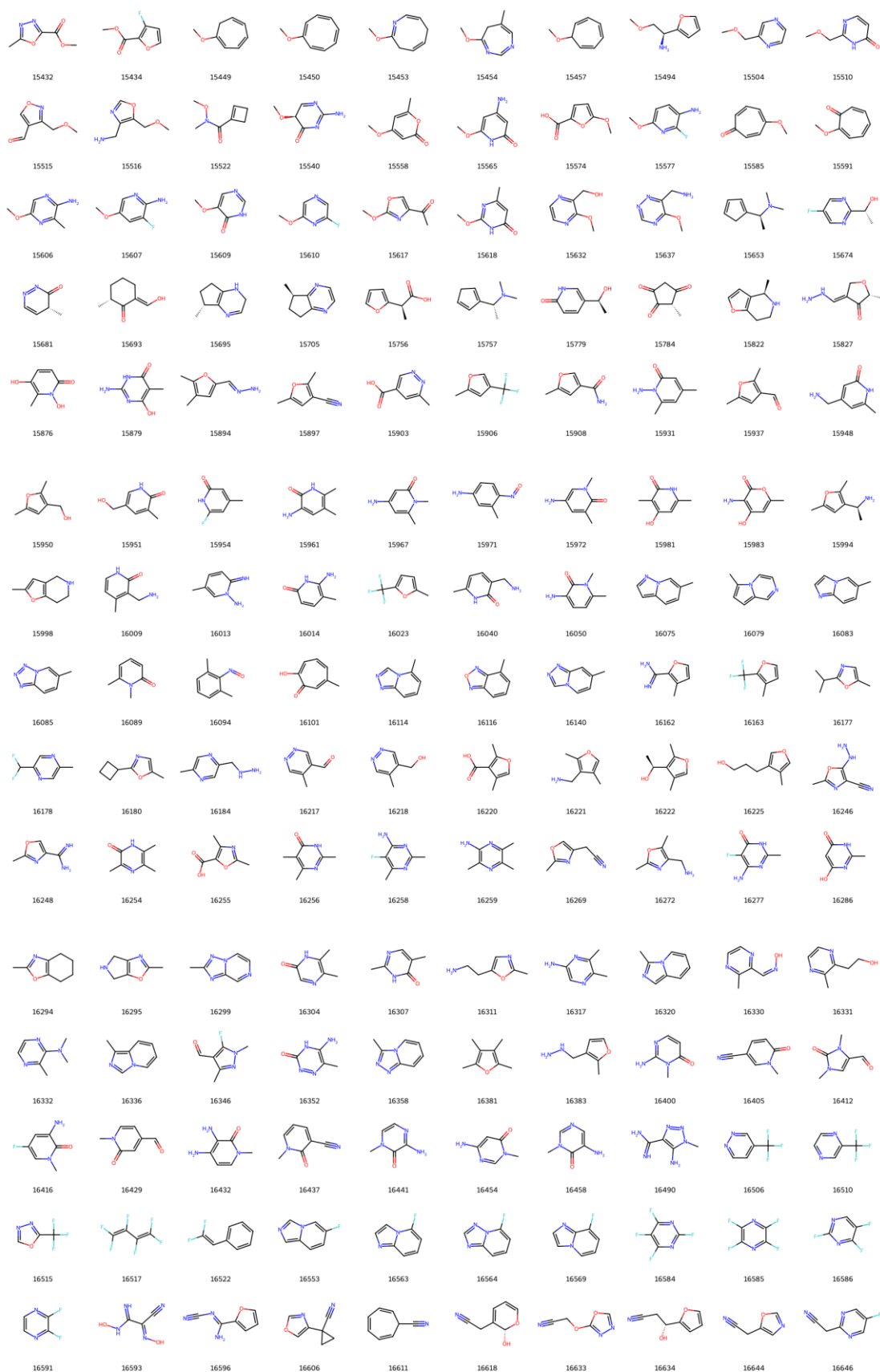
Section Detail of All Selecting Reactants

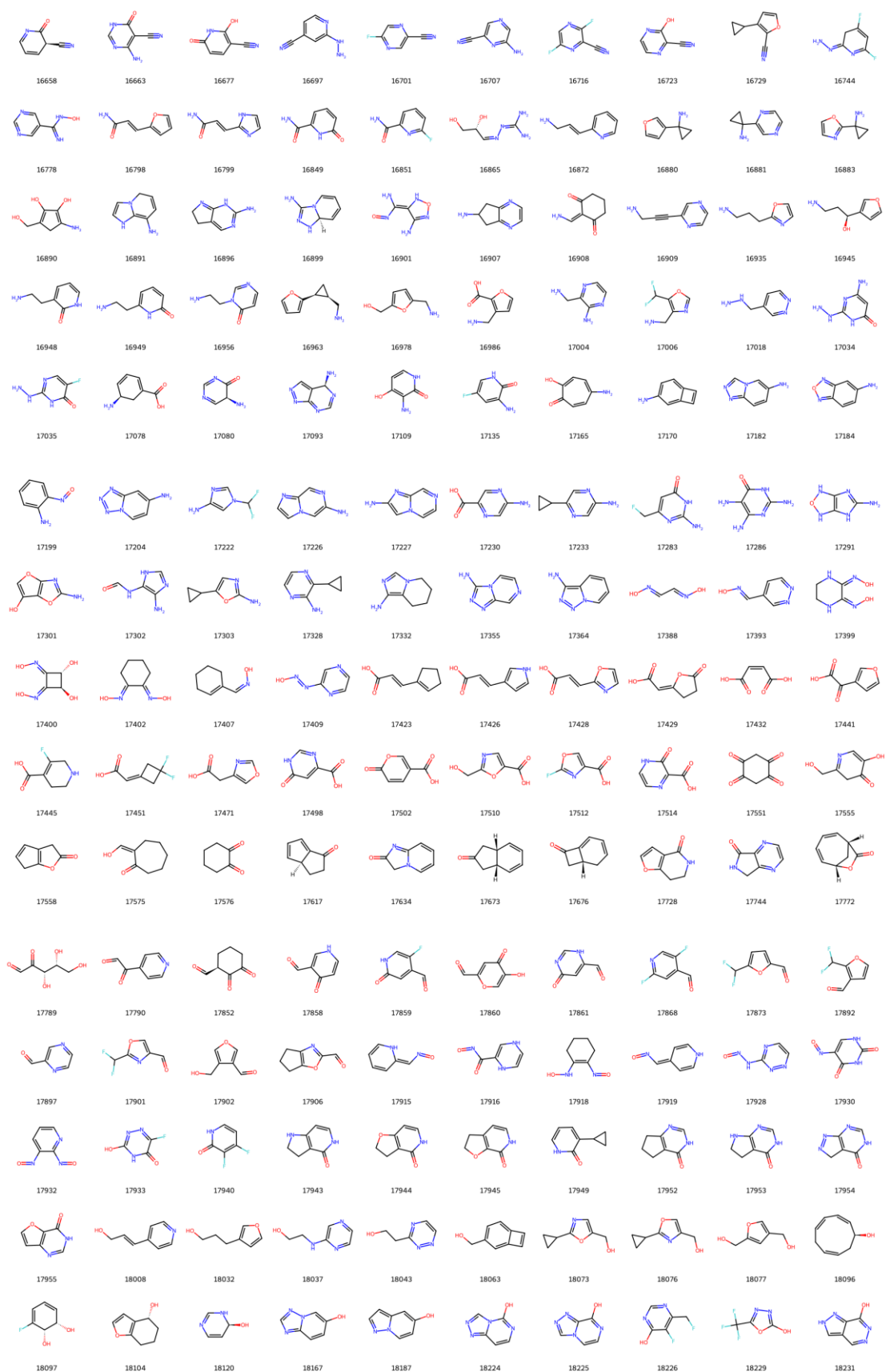
All selected 4π are showed as follows, and the id of each molecules is listed bellow the molecules.

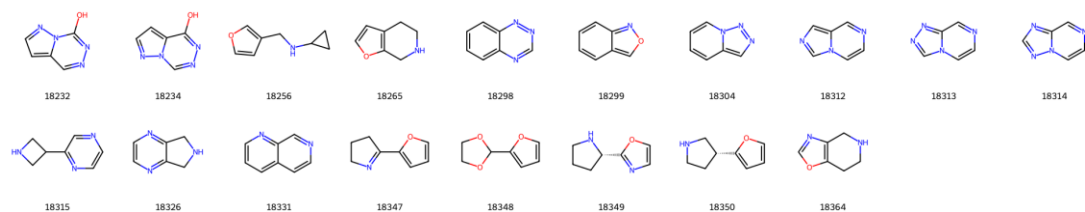












And all selected 2π are showed as follows:

